C/C++ Operator Precedence & Associativity

A beginner-friendly guide with simple examples. The main idea is: **higher precedence = higher priority**.

1. What is Operator Precedence?

When an expression contains several operators, precedence tells the compiler which operator should be handled first.

Example:

```
int x = 2 + 3 * 4;
```

Multiplication has higher precedence than addition, so we do 3 × 4 first.

```
2 + (3 * 4) = 2 + 12 = 14
```

2. Beginner Precedence List

Priority	Operators	Meaning	Associativity	
1	()	Parentheses / function call	Left → Right	
2	++ -- ! ~	Unary operators	Right → Left	
3	* / %	Multiply / divide / remainder	Left → Right	
4	+ -	Add / subtract	Left → Right	
5	<< >>	Bitwise shift	Left → Right	
6	< <= > >=	Relational comparison	Left → Right	
7	== !=	Equality comparison	Left → Right	
8	&	Bitwise AND	Left → Right	
9	^	Bitwise XOR	Left → Right	
10	|	Bitwise OR	Left → Right	
11	&&	Logical AND	Left → Right	
12	||	Logical OR	Left → Right	
13	?:	Ternary	Right → Left	
14	=	+= -= ...	Assignment	Right → Left
15	,	Comma	Left → Right	

3. What is Associativity?

If two operators have the **same precedence**, associativity tells us which direction to follow.

Left → Right

```
int x = 20 / 5 * 2;
```

Both / and * have the same precedence, so evaluate from left to right:

```
(20 / 5) * 2 = 4 * 2 = 8
```

Right → Left

```
a = b = 10;
```

Assignment works from right to left:

```
a = (b = 10)
```

First b becomes 10, then a becomes 10.

4. More Simple Examples

Example 1: Multiplication before Addition

```
int x = 5 + 2 * 3;
```

Result: 5 + (2 * 3) = 5 + 6 = 11

Example 2: Parentheses have priority

```
int x = (5 + 2) * 3;
```

Result: (5 + 2) * 3 = 7 * 3 = 21

Example 3: Same precedence

```
int x = 10 - 4 + 2;
```

Result: (10 - 4) + 2 = 8

Example 4: Division and multiplication

```
int x = 20 / 5 * 2;
```

Result: (20 / 5) * 2 = 8

Example 5: Shift

```
int x = 5 + 2 << 1;
```

Result: (5 + 2) << 1 = 7 << 1 = 14

Example 6: Comparison

```
int x = 5 + 2 > 6;
```

Result: (5 + 2) > 6 → 7 > 6 → true (1)

Example 7: Logical operators

```
int x = 5 > 3 && 2 > 1;
```

Result: (5 > 3) && (2 > 1) $\rightarrow$ 1 && 1 $\rightarrow$ true

5. Bitwise Operators and Precedence

For bitwise operators, remember that shifts come before comparisons, while bitwise AND/XOR/OR come after equality comparisons.

Example: Left Shift

```
int x = 5 << 2;
```

5 = 00000101■
5 << 2 = 00010100■ = 20

Example: NOT

```
~5
```

Using 8 bits: 00000101 → 11111010. In C/C++, with two's-complement signed integers, ~5 = -6.

6. The Most Important Rules to Memorize

For beginner C programming, first memorize this smaller list:

```
()
* / %
+ -
< <= > >=
== !=
&&
||
=
```

Remember: Higher in the list means higher priority.

7. Quick Practice

1. int x = 2 + 3 * 4; → What is x?

2. int x = (2 + 3) * 4; → What is x?

3. int x = 20 / 5 * 2; → What is x?

4. int x = 10 - 3 + 2; → What is x?

5. int x = 5 + 2 << 1; → What is x?

6. a = b = 7; → Which assignment happens first?

Practice Answers

1. 14

2. 20

3. 8

4. 9

5. 14

6. b = 7 happens first, then a = 7